EE6083 VLSI Testing—Homework 3

1. **Scan Test**
   Figure 1 shows the circuit of a FSM. Complete the following questions.

   (a) Convert the circuit by adding scan to the three flip-flops. Create a complete scan test for the indicated faults (α, β, γ). Show the sequence of test vectors that are applied to this circuit in order to detect the faults, and show the sequence required to scan out and observe the results. (3 points)

   (b) Think an approach to reduce the switching power induced by the shifting operation of the scan without incurring extra hardware overhead. (3 points)

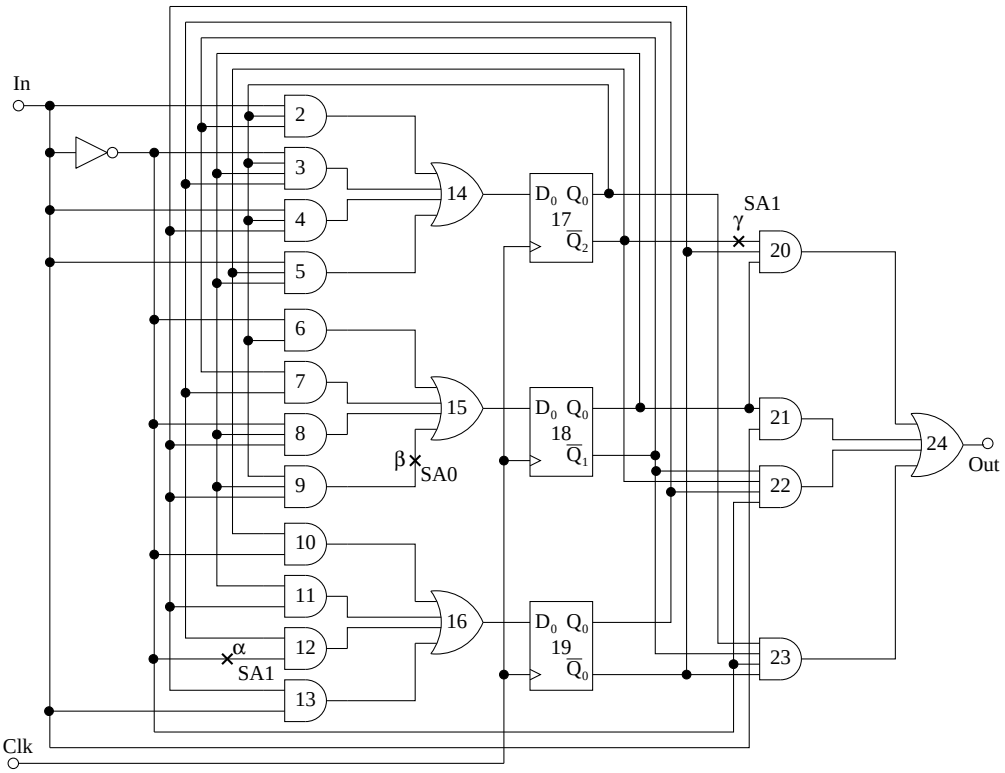

Figure 1: An example of a FSM.

2. **Scan Testing Time Reduction**
   A circuit with 1000 scan flip-flops has three uniquely identifiable blocks of logic. Block1 has 200 scan flip-flops and requires 300 scan vectors, block2 has 300 flip-flops and requires 80 scan vectors, and block3 has 500 scan flip-flops and requires 700 vectors.

   (a) If you could not break up the scan chains, how would you organize them to minimize test time? (2 points)

   (b) If you could break up the longest scan chain into scan chains of length 300 and 200, with each requiring 400 scan vectors, how would you organize the scan chains to minimize test time? (3 points)

3. **LFSR**
   Design a modified 6-bit LFSR to generate exhuastive test patterns, i.e., 64 different test patterns. Implement your design using the Verilog HDL and show the simulation waveforms to justify your design. (4 points)